• Objective:

1. Design a buck-converter w/ specific input & output specs

2. Needs a small-signal model of the converter

3. Design a Type-II Compensator for closed-loop stability

4. Simulate the system and analyze performance

5. Implement the design in hardware

6. Validate results

• Specifications:

1/ Input Voltage: 12V

2/ Output Voltage: 5V

3/ Output Current: 2A

4/ switching frequency: 100 kHz

5/ Output Ripple: < 50mV

6/ Efficiency: >85%.

7/ Phase Margin: 45°-60°

• System Overview:

- Power Stage (MOSFET, Diode, Inductor, apacitor)
- PWM Control Block
- Feedback Network
- Type - II Compensation Network

Values:

// Will adjust later

- f (switching frequency): 100 KHz
- D (Duty Cycle): 0.416 = **41.6%**
- Δi_L (Ripple current @ inductor): 2.5 A
- $i_{L,min}$: 0.75 A
- $i_{L,max}$: 3.25 A
- L (inductor): ~11.68 μH
- C (capacitor): ~62.5 μF

Power stage

- Switching Frequency:

//Duty Ratio

f, switching frequency

$$D = \frac{t_{on}}{T} = t_{on} \cdot f$$

Note:

- Frequency influences the power loss in switching

$\uparrow f \Rightarrow \uparrow$ Power loss

$P = V * I$

$$\Delta i_L = \left(\frac{V_s - V_o}{L}\right) DT = \frac{V_s - V_o}{L \cdot f}$$

$$L = V_o \frac{(1-D)}{\Delta i_L \cdot f}$$

to avoid any audio noise

- Switch frequency is usually above 20kHz & between 100kHz – 1MHz
- Some designers choose 500kHz (middle ground)

$$\text{Duty Ratio} = \frac{t_{on}}{T} = t_{on} \cdot f = t_{on} \cdot 500\,kHz$$

Duty Cycle:

$$D_{cycle} = \frac{V_{out}}{V_{in}} = \frac{5\,V}{12\,V} = 0.416 \approx 41.6\,\%$$

Ripple current/Inductor:

// choose a ripple current to determine the inductor

$$L = \frac{V_o (1 - D)}{\Delta i_L \cdot f}$$

$$L = \frac{5v\,(1 - 0.416)}{2.5A\,(100kHz)} = 1.168 \times 10^{-5} = 11.68\,\mu H$$

// To keep a stable circuit, L_{min} MUST be less than the inductor value

$$I_{min} = I_L - \frac{\Delta i_L}{2} = 2A - \frac{2.5A}{2} = 2 - 1.25 = \boxed{0.75A}$$

// $I_L = I_R = I_o$ because Since it is a steady state current through the inductor is the same as current through the resistor

// It is in steady state because the current through the capacitor is zero over the average of one switching period

// It is NOT instantaneously $I_L = I_R$ but through average time $I_L = I_R$

$$I_{max} = I_L + \frac{\Delta i_L}{2} = 2 + 1.25 = \underline{3.25A}$$

// Now we have to confirm the ripple current chosen was acceptable

// We assume:

- $i_L(t) > 0$, for the entire cycle so it's Continuous Conduction Mode (CCM)
- We force $I_{min} = 0$ to find minimum inductance that keeps current continuous
- Finally, use $\Delta i_{L,calc} = 2I_L$ to get the maximum ripple allowed to be in CCM

// If $\Delta i_{L,chosen} < i_{L,cal} = 2I_L$, then the converter operates in CCM

$$\Delta i_{L,chosen} = 2.5A$$

$$I_{min} = 0 = I_L - \frac{\Delta i_{L,calc}}{2}$$

$$\Delta i_{L,cal} = 2I_L$$

$$\Delta i_{L,cal} = 2(2A) = 4A$$

$$\underline{2.5A < 4} \checkmark$$

// 2.5A as the ripple current is acceptable because it is below the margin and provides some design margin

Output Capacitor / Voltage ripple:

$$\Delta Q = \frac{1}{2}\left(\frac{T}{2}\right)\left(\frac{\Delta i_L}{2}\right) = \frac{T\Delta i_L}{8}$$

$$Q = C \cdot V_o \Rightarrow \Delta Q = C \cdot \Delta V_o$$

$$\Delta i_L = \frac{V_o(1-D)}{Lf}$$

$$C \cdot \Delta V_o = \frac{T \cdot \Delta i_L}{8}$$

$$\Delta V_o = \frac{T \cdot V_o (1-D)}{8LfC} = \frac{V_o(1-D)}{8LCf^2}$$

$$C = \frac{V_o(1-D)}{8Lf^2 \Delta V_o}$$

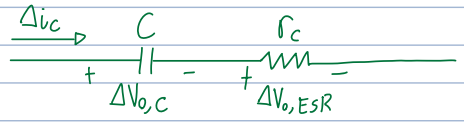
$$C = \frac{5\ (1 - 0.416)}{8(11.68\mu H)(100kHz)^2(50mV)}$$

$$C = 62.5\mu F$$

// This the smallest capacitance we can go so $\Delta V_o < 50mV$

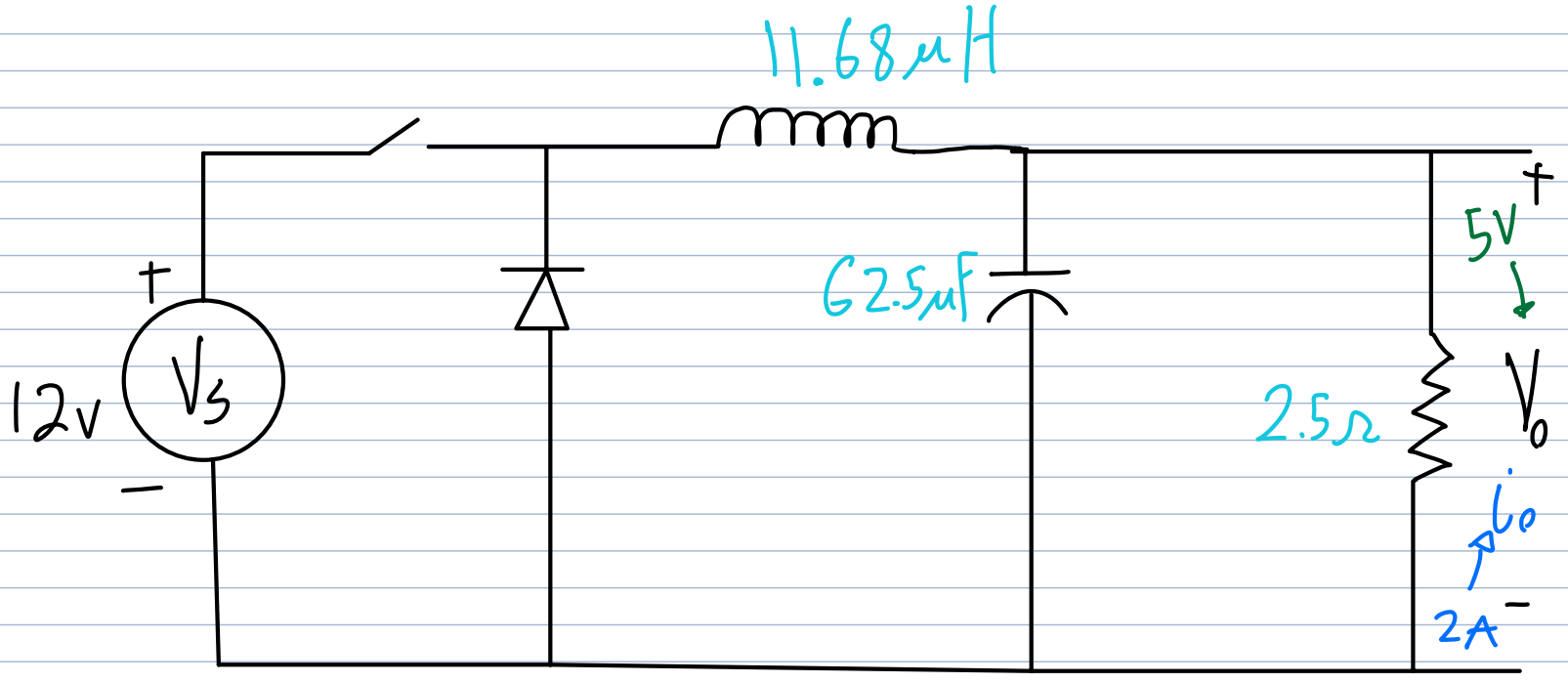
// Side Note:

- Capacitor ESR (Equivalent Series Resistance) can significantly effect the output voltage ripple
- Larger capacitance results in a lower ESR
- $\Delta V_o \approx \Delta V_{o,ESR} = \Delta i_C r_C = \Delta i_L r_C$
- $i_C = i_L - i_R$

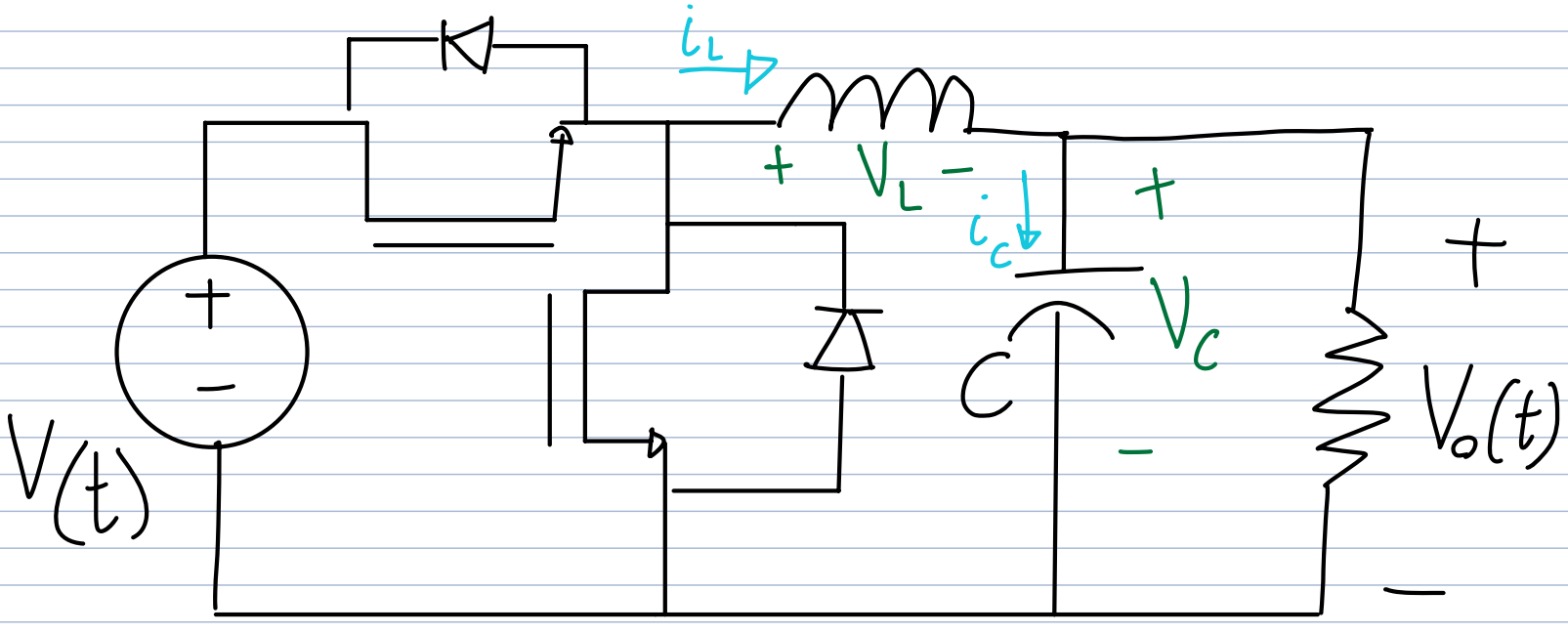


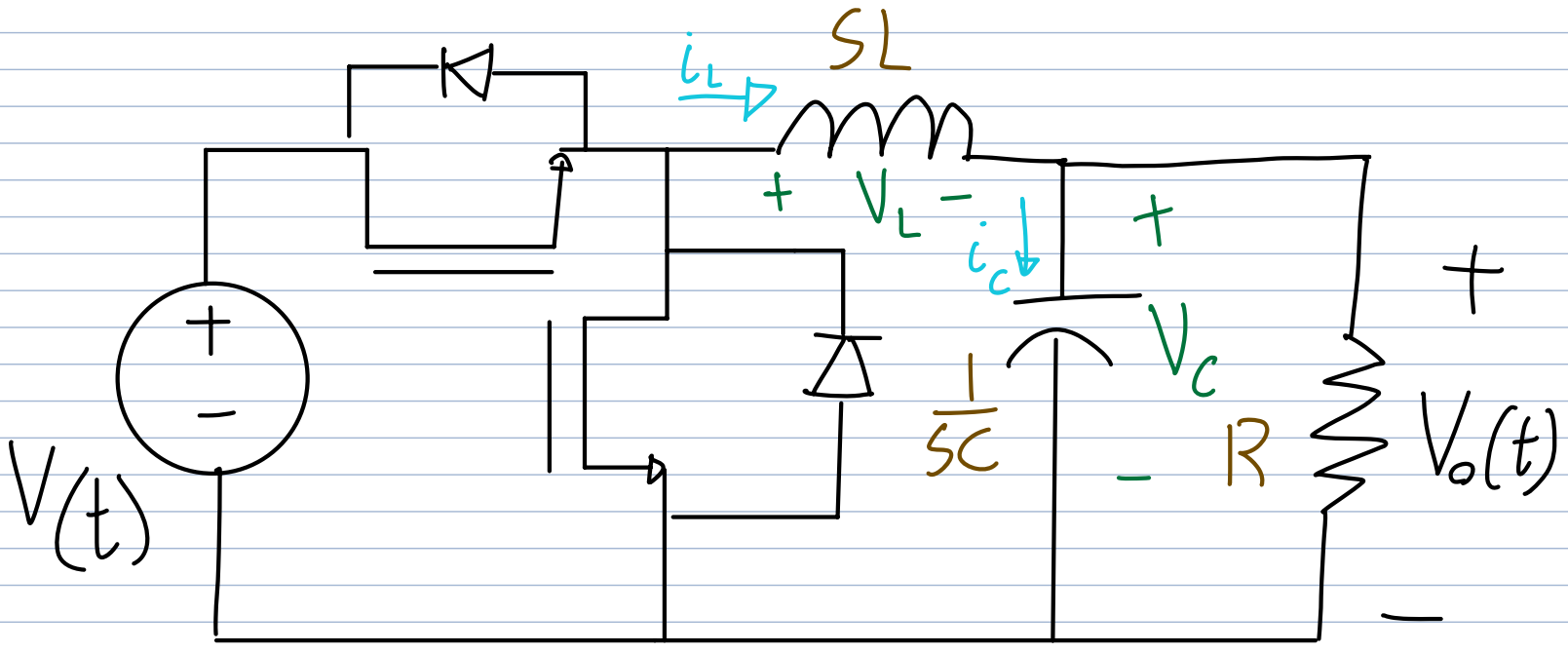
Modeling and Analysis

Derive Control-to-Output transfer function



$$R = \frac{V}{I} = \frac{5}{2} = \boxed{2.5\Omega}$$





// "Control-to-Output" means the variable that controls our possible output
// In this case Duty-cycle (which we control) is what can determine our output (V_o)
// we don't control impedance [$Z(s)$]
// We don't control the gain b/c it's constant [$V_g(s)$]

(1) $G_{Vd}(s) = \frac{\tilde{V}_o(s)}{\tilde{d}(s)}$ (2) $V_x(t) = d(t) \cdot V_s \Rightarrow \frac{V_x(t)}{d(t)} = V_s$

(3) $d(t) = D + \tilde{d}(t) = 1$

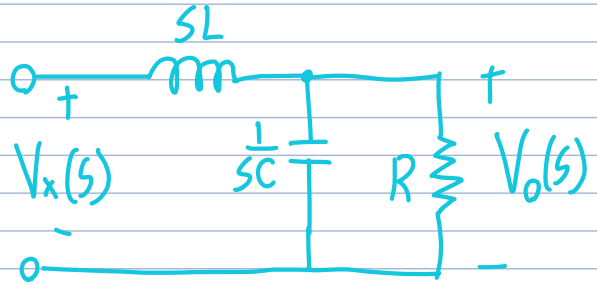
// combine (2) + (3)

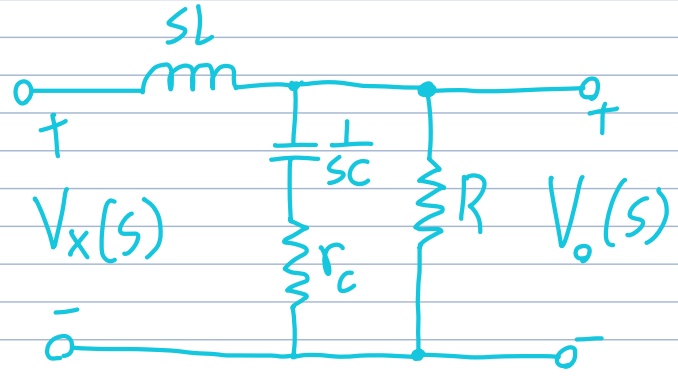
// Small signal form

(2) + (3)

$V_x(t) = (D + \tilde{d}(t)) \cdot V_s \Rightarrow \boxed{D \cdot V_s + \tilde{d}(t) \cdot V_s}$

LC Filter:





(1) $V_x - V_o = L\frac{di_L}{dt}$ // Laplace transformation

$V_x - V_o = L(s \cdot I_L)$

(2) $i_L - \frac{V_o}{R} = C\frac{dV_o}{dt}$ // Laplace transformation

$I_L - \frac{V_o}{R} = C(s \cdot V_o)$

// Now isolate I_L @ (2)

$I_L = sC \cdot V_o - \frac{V_o}{R}$ // Now Plug I_L to equation (1)

(1)+(2)

$V_x - V_o = sL\left[sC \cdot V_o + \frac{V_o}{R}\right]$

$V_x - V_o = s^2LC \cdot V_o + s\frac{L \cdot V_o}{R}$

$V_x = s^2LC \cdot V_o + s\frac{LV_o}{R} + V_o$

$$\frac{V_o}{V_x} = \frac{1}{s^2LC + s\frac{L}{R} + 1}$$

// Now Combine w/ switch function $\left(\frac{\tilde{V}_x}{\tilde{d}_x} = V_s\right)$

$$G_{vd}(s) = \frac{\tilde{V}_o}{\tilde{d}} = \frac{\tilde{V}_o}{\tilde{V}_x} \cdot \frac{\tilde{V}_x}{\tilde{d}x}$$

$$G_{vd}(s) = \left(\frac{1}{s^2 LC + s\frac{L}{R} + 1}\right)(V_s)$$

$$\boxed{G_{vd}(s) = \left(\frac{V_s}{s^2 LC + s\frac{L}{R} + 1}\right)}$$

Identify poles and zeros

$$G_{vd}(s) = \left(\frac{V_s}{s^2 LC + s\frac{L}{R} + 1}\right)$$

$$\Rightarrow \frac{12\text{V}}{s^2(11.68\mu\text{H})(62.5\mu\text{F}) + s\frac{11.68\mu\text{H}}{2.5\Omega} + 1}$$

$$= \frac{12\text{V}}{(7.30\times10^{-10})s^2 + (4.672\times10^{-6})s + 1}$$

// No zeros

// Find Poles Now

$$\frac{1}{\frac{1}{LC}\left(s^2 LC + s\frac{L}{R} + 1\right)} \Rightarrow \frac{1}{s^2 + s\frac{1}{RC} + \frac{1}{LC}}$$

// To isolate s^2

Poles:

- $s^2 + s\frac{1}{RC} + \frac{1}{LC}$

$$\frac{1}{RC} = \frac{1}{(2.5)(62.5\times10^{-6})} = 6400$$

$$\frac{1}{LC} = \frac{1}{(11.68\times10^{-6})(62.5\times10^{-6})} = 1.37\times10^{9}$$

$$\therefore s^2 + 6400s + 1.37\times10^{9} = 0$$

// quadratic formula

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-(6400) \pm \sqrt{(6400)^2 - 4(1)(1.37\times10^9)}}{2(1)}$$

$$s = \frac{-6400 \pm \sqrt{-5.43904\times10^9}}{2}$$

$$s = \frac{-6400 \pm j\sqrt{5.43904\times10^9}}{2}$$

$$s = \frac{-6400 \pm j\,73749}{2}$$

$$\boxed{s_{1,2} = -3200 \pm j\,36874.5}$$

// Two Poles

// resonant frequency

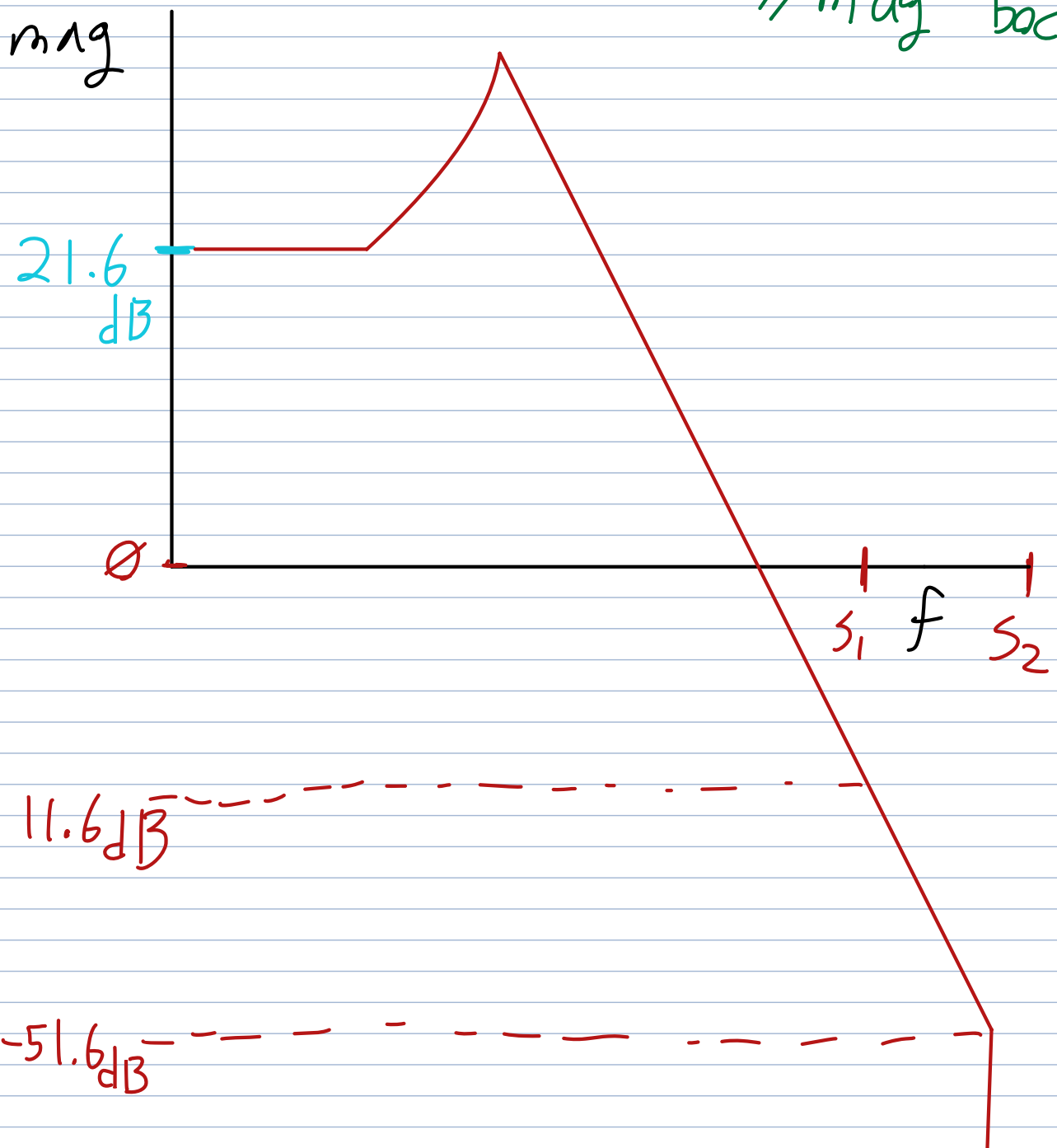
$$\omega_0 = \frac{1}{\sqrt{LC}} = \boxed{3.70\times10^4 \text{ rad/s}}, \quad f_0 = \frac{\omega_0}{2\pi} \simeq \boxed{5.9\text{k Hz}}$$

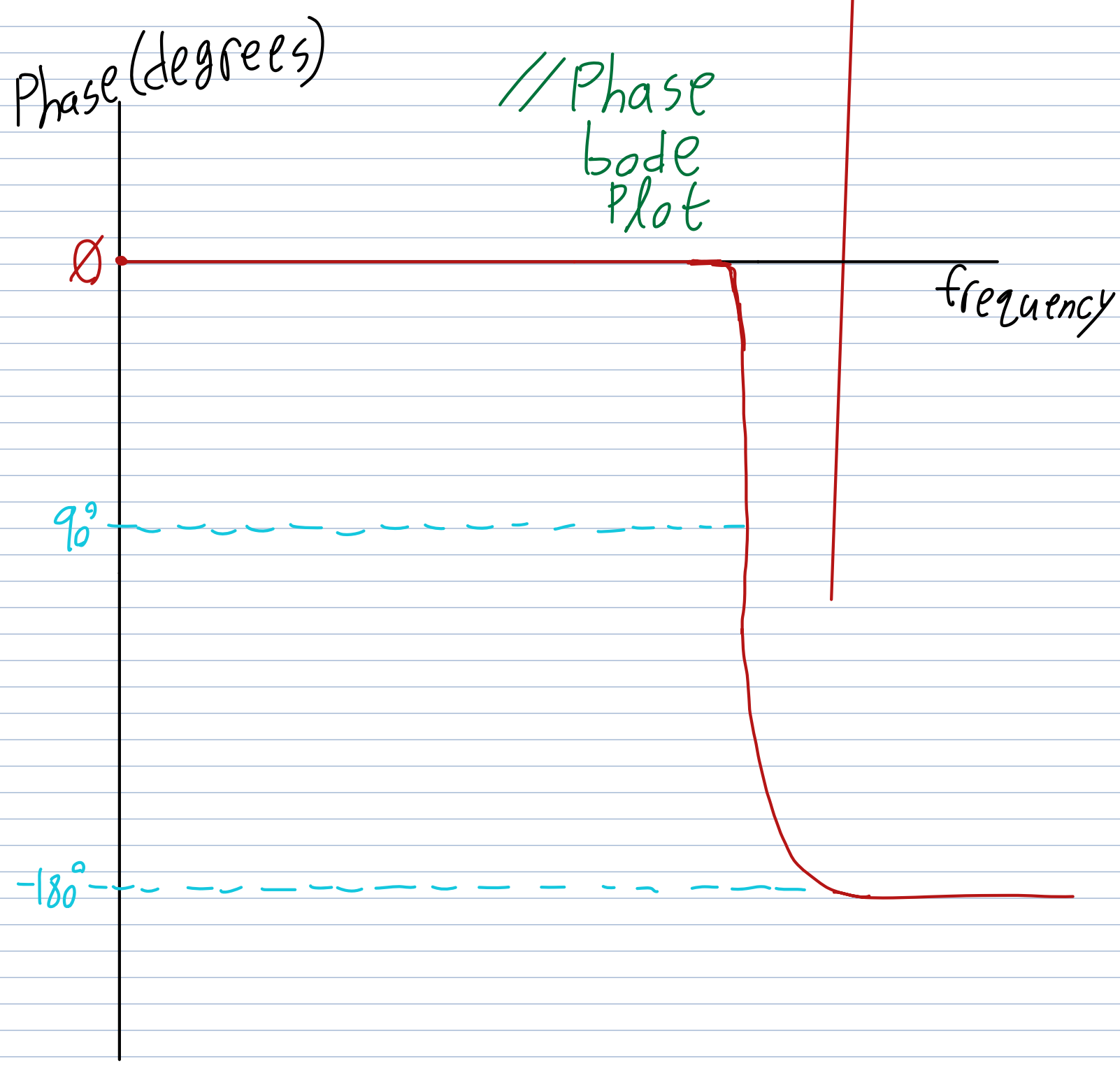
//DC Gain

$$|G_{Vd}(0)| = 12 \Rightarrow 20\log_{10}(12) \simeq 21.6\ dB$$

Generate bode Plot

//mag bode plot





//To use the python code Jahan provided, I need to find V_{ramp} & R_{ESR}

//Value also comes from PWM block

- Assume $V_{ramp} = \boxed{1.0\ V} = \frac{d(s)}{V_c(s)} = \frac{1}{V_p}$

//V_p = Sawtooth amplitude

- $\Delta V_{o,ESR} = \Delta i_c \cdot r_c$

$\Rightarrow r_c = \frac{\Delta V_{o,ESR}}{\Delta i_c} = \frac{50mV}{2.5} = \boxed{0.02\ \Omega}$

//MAX R_{ESR}

$R_{ESR} \leq 20mV$

Type - II Compensation Design

Add integrator (Pole @ Origin)

// The point of these Compensator designs are to shape the frequency response of the system by adding phase boost near the crossover frequency, ensuring that the phase margin is sufficient and the system remains stable

// In other words it:

- Increases low-frequency gain
  - Reduces steady-state error
- Add Phase near cross over
  - improves stability
- Reduce high-frequency gain
  - filters switching noise

// What is crossover?

- Gain Crossover (ω_c):
  - @ Gain = 1 (0dB)
- Phase Crossover:
  - @ Phase = -180°

// We want the Gain to be @ 1 (0dB), Phase > -180, and P.M. (Phase Margin) > 45° but PM < 60°

$$f_c = \frac{f_{sw}}{10} = \frac{100k}{10} = \boxed{10kHz}$$

// The cross over frequency for Gain crossover is @ 1 (0dB)